IIIT KOTTAYAM · DEPARTMENT OF CSE & MATHEMATICS AND COMPUTING

Presentation Delivery Scripts (10-Minute Team Script)

Course: IEG313: Quantum Computing for Engineers **Team:** Dirac's Army (Group 5)

Evaluators: HOD (CSE) & HOD (Math & Computing) **Project:** QKernel-Lite

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🎙️ QKernel-Lite: Complete Team Presentation Scripts

Course: IEG313: Quantum Computing for Engineers · IIIT Kottayam
Team: Dirac's Army (Group 5)
Total Target Duration: 9 minutes 30 seconds (Budget: 9–10 minutes)
Evaluators: Head of Department (Computer Science & Engineering) & Head of Department (Mathematics & Computing)

⏱️ Master Timing & Stage Coordination Table

Presenter	Assigned Modules	Slides	Target Time	Cumulative Time
Daksh Singh	Title, Thesis, Data Scarcity, Paired t-Test, Bonferroni	01, 02, 03, 04, 05	**2 min 00 s**	0:00 – 2:00
Anshuman Biswas	Scratch CNN (M1), Frozen ResNet-18, PCA, Classical Control (M2)	06, 07	**1 min 50 s**	2:00 – 3:50
Himmat Sirwee	8Q Angle Encoding, Statevector Sim, 1-Layer Closed Form, Small-s Duality	08, 09	**1 min 50 s**	3:50 – 5:40
Sawan Kumar	Phase Wrapping Bug, PQK vs. Fidelity Plateaus, ZZ Correlators	10, 11, 12	**2 min 00 s**	5:40 – 7:40
Aryan Gupta	960-Run Ledger, Budget Asymmetry, Prototype Platform, Bounded Conclusions	13, 14, 15, 16	**2 min 05 s**	7:40 – 9:45

1. Daksh Singh (2024BCS0042)

Role: Opening, Problem Definition, Dataset Protocol, Statistical Framework
Assigned Slides: 01, 02, 03, 04, 05
Allotted Time: ~2 minutes 00 seconds (~330 words)

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slide-1 | Slide 01: Title & Cover

(Target Time: ~15 seconds)

> **[Spoken Delivery]**
> *"Respected Heads of Departments, Faculty, and colleagues, good morning.*

We are Dirac's Army, and today we present QKernel-Lite: an empirical, paired benchmark evaluating Projected Quantum Kernel Support Vector Machines against classical transfer learning baselines under extreme medical data scarcity."

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slide-2 | Slide 02: The Project in Six Incontrovertible Scientific Facts

(Target Time: ~35 seconds)

> **[Spoken Delivery]**
> *(Point to Left Card — Methodological Framework)*
> *"To orient our defense, our entire study reduces to six rigorous findings.*

Simply put: recent literature hypothesizes that mapping classical data into quantum Hilbert spaces grants an inductive advantage when training labels are scarce. To test this honestly, we took pediatric chest X-rays with sample sizes as small as 10 to 100 scans, extracted 8 clinical features via frozen ResNet-18, and compared classical Gaussian RBF kernels against projected quantum kernels using the exact same convex dual quadratic solver.

(Point to Right Card — Measured Scientific Reality)
The measured reality was unambiguous: the classical model won decisively across every sample size. In fact, quantum entanglement did not create advantage—it collapsed feature variance by 34-fold and systematically degraded diagnostic performance. The only quantum circuit matching classical accuracy was a 1-layer unentangled product state, which is trivially classically simulable."

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slide-3 | Slide 03: Module Ownership & Presentation Sequence

(Target Time: ~15 seconds)

> **[Spoken Delivery]**
> *(Glance at Architecture Diagram)*
> *"To ensure complete transparency, every member of our team independently designed, implemented, and defends their assigned module—from statistical discipline and deep feature extraction to quantum circuit mathematics and platform engineering."*

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slide-4 | Slide 04: Pediatric Radiography & Extreme Label Scarcity

(Target Time: ~25 seconds)

> **[Spoken Delivery]**
> *(Point to Left Box — Experimental Protocol)*
> *"Our clinical testing ground is PneumoniaMNIST, consisting of pediatric chest radiographs. In rare pediatric illnesses or novel outbreaks, clinicians often have fewer than 20 verified scans.*

We simulated four extreme scarcity regimes: $N = 10, 20, 50,$ and 100 balanced training scans. But crucially, look at the right card: we maintained an uncompromised, fixed evaluation set of 624 real patient scans (234 normal, 390 pneumonia).

Why ROC-AUC instead of raw accuracy? Because the test set is 62.5% pneumonia. A naive model predicting pneumonia every single time scores 62% accuracy with zero diagnostic sensitivity. ROC-AUC evaluates true ranking invariant to class imbalance."

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slide-5 | Slide 05: Paired Student's t-Test & Bonferroni Family-Wise Control

(Target Time: ~30 seconds)

> **[Spoken Delivery]**
> *(Point to Left Equation — Paired Student's t)*
> *"To ensure statistical discipline, we ran a fully paired experimental design across 15 independent random seeds ($df = 14$). By evaluating all models on the identical patient subsets per seed, between-sample variance cancels out, isolating true model deltas.*

With $n = 15$, population variance is unknown, requiring a Student's t-distribution where $t_{0.025,14} = 2.145$. Furthermore, because we conducted 12 primary pairwise comparisons, standard $\alpha = 0.05$ guarantees false positives. We enforced a strict Bonferroni correction: $\alpha_{\text{bonf}} = 0.05/12 = 0.00417$.

(Point to Right Card — Equivalence Gate)
Every single quantum deficit survived this threshold ($p < 0.001$). Furthermore, under a strict clinical equivalence interval of ± 0.02 AUC, entangled circuits failed catastrophically.

Now, I hand over to Anshuman to walk through our deep classical baselines."

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2. Anshuman Biswas (2024BCD0008)

Role: M1 (Scratch CNN), ResNet-18 Transfer Feature Pipeline, PCA Low-Rank Reduction, Model M2 (Classical Control)
Assigned Slides: 06, 07
Allotted Time: ~1 minute 50 seconds (~320 words)

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slide-6 | Slide 06: M1 Scratch CNN & Frozen ResNet-18 Transfer Learning

(Target Time: ~55 seconds)

> **[Spoken Delivery]**
> *"Thank you, Daksh. Good morning Professors.*

To benchmark quantum kernels fairly, we first need to know what classical deep learning actually does under extreme scarcity.

(Point to Left Card — M1 Scratch CNN)
Our first baseline, Model M1, is a standard convolutional neural network trained from scratch. In plain terms: when an emergency clinic only has 10 X-rays, training 105,000 weights is like asking a student to memorize an entire textbook after seeing only two practice questions. Mathematically, our parameter-to-data ratio exceeds $10,000$ to 1 ($P/N > 10^4$). Under such severe parameter under-determination, gradient descent easily overfits scanner artifacts and bedframe shadows.

Yet, notice an honest empirical finding: M1 did not collapse to random guessing—it achieved 0.847 AUC at $N = 10$. Pneumonia opacities produce large, distinct optical consolidations that simple spatial convolutions can partially capture.

(Point to Right Card — Frozen ResNet-18)
To solve overfitting properly, we used Transfer Learning. Instead of training on 10 scans, we freeze an 11.7-million parameter ResNet-18 pretrained on ImageNet. Why freeze it completely? Because fine-tuning 11 million weights on 10 scans guarantees catastrophic representation collapse. By locking the weights and fixing batch-normalization in `eval()` mode, its convolutional primitives—edges, lung textures, and tissue boundaries—transfer directly to chest radiographs. Passing each scan through Global Average Pooling extracts a rich, noise-free 512-dimensional latent vector."

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slide-7 | Slide 07: PCA Dimensionality Reduction & Classical Control (M2)

(Target Time: ~55 seconds)

> **[Spoken Delivery]**
> *(Point to Left Card — PCA Compression)*
> *"Now comes the mathematical bridge between our classical features and quantum simulation.*

Simulating an n-qubit register scales as 2^n. Simulating 512 qubits is physically impossible on classical hardware, so we must compress 512 dimensions to 8 principal components.

Here, we enforced strict linear algebra discipline: PCA is fitted exclusively on the training fold. At $N = 10$, the sample covariance matrix has a rank of at most $N - 1 = 9$. We extract the top 8 eigenvectors—retaining 77% of cumulative variance—and min-max scale the coordinates to $[0, \pi]$ so they can drive physical quantum rotation angles θ_i. Crucially, test scans are projected using training fold eigenvectors and min-max boundaries, guaranteeing zero data leakage.

(Point to Right Card — Model M2 RBF-SVM)
This establishes Model M2, the scientific linchpin of our entire research capstone.

In published literature, authors frequently compare quantum models against unregularized scratch CNNs and claim 'quantum advantage'. That is an artificially weak strawman. Model M2 consumes the exact same 8 PCA components and maps them through a classical Gaussian RBF kernel into an identical dual convex quadratic programming SVM solver.

The result? Classical transfer learning provides a massive diagnostic lift: $+0.0488$ *AUC at* $N = 10$*, and a statistically verified* $+0.0445$ *lift at* $N = 20$ *(*$p = 0.0010$*, surviving Bonferroni). This is the true benchmark quantum must beat.*

Now, Himmat will explain how these 8 clinical angles are encoded into quantum circuits."

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3. Himmat Sirwee (2024BCS0040)

Role: Quantum Register Topology, Statevector Simulation, 1-Layer Closed Form, Small-s Taylor Duality
Assigned Slides: 08, 09
Allotted Time: ~1 minute 50 seconds (~310 words)

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slide-8 | Slide 08: 8-Qubit Angle Encoding & Circuit Topology

(Target Time: ~55 seconds)

> **[Spoken Delivery]**
> *"Thank you, Anshuman. Good morning Professors.*

(Point to Left Card — Circuit Topology)
To build our quantum pipeline, we initialize an 8-qubit register in the ground state $|00000000\rangle$, where each qubit corresponds to one compressed clinical feature coordinate $\theta_i \in [0, \pi]$.

On each qubit i, we apply a rotation block $R_Y(s\theta_i)$ followed by $R_Z(s\theta_i)$, where s is an encoding scale parameter. In entangled architectures, we interleave these rotations with a circular CNOT chain—connecting qubit 0 to 1, 1 to 2, and wrapping 7 back to 0. In 2-layer circuits, this entire block is repeated twice.

(Point to Right Card — Simulation vs. Physical Hardware)
Now, look at the simulation methodology on the right. We execute exact statevector simulation using PennyLane's `lightning.qubit` engine. An 8-qubit statevector requires $2^8 = 256$ complex amplitudes, which is classically simulated without Monte Carlo shot noise.

This is a crucial scientific distinction: physical NISQ QPUs suffer from thermal relaxation (T_1), dephasing (T_2), and finite measurement sampling error. Because our statevector simulation has zero physical hardware noise, any deficit we observe in quantum performance is purely algorithmic and mathematical, not an artifact of noisy hardware."

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slide-9 | Slide 09: Mathematical Derivation: 1-Layer Closed Form & Small-s Duality

(Target Time: ~55 seconds)

> **[Spoken Delivery]**
> *(Point to Left Box — Exact Closed Form on Slide 9)*
> *"To understand what the quantum circuit is doing geometrically, we derived the analytical closed form for the 1-layer unentangled product state shown on Slide 9.*

Applying $R_Y(s\theta)$ to $|0\rangle$ rotates the state in the X-Z plane, and $R_Z(s\theta)$ injects a relative phase shift $e^{is\theta}$ into the $|1\rangle$ component.

By calculating the expectation values of the Pauli operators, we obtain the exact coordinates of the Bloch vector shown in the box: $\langle X\rangle = \sin(s\theta)\cos(s\theta)$, $\langle Y\rangle = \sin^2(s\theta)$, and $\langle Z\rangle = \cos(s\theta)$.

(Point to Right Box — Taylor Duality)
Now examine the right box: what happens when the scale factor s is tuned small ($s \leq 0.125$)?

Taking the first-order Taylor series around zero reveals a profound duality: $\langle X\rangle$ becomes approximately $s\theta$, while $\langle Y\rangle$ and $\langle Z\rangle$ become quadratic in $(s\theta)^2$.

Notice that $\langle X \rangle$ is strictly linear in the input feature, while $\langle Y \rangle$ and $\langle Z \rangle$ are quadratic. When these 24 expectation values feed an RBF kernel, the quantum model mathematically reduces to a classical polynomial approximation!

This proves why the quantum model performs well at small s: it recovers classical accuracy precisely because its quantum characteristics have been suppressed to mimic a classical kernel.

Now, Sawan will demonstrate what happens when default quantum parameters break down."

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4. Sawan Kumar (2024BCS0164)

Role: Phase Wrapping Bug, PQK vs. Fidelity Plateaus, Metric Diagnostics, Refuted ZZ Correlator Hypothesis
Assigned Slides: 10, 11, 12
Allotted Time: ~2 minutes 00 seconds (~330 words)

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slide-10 | Slide 10: Physical Failure Modes: Phase Wrapping & Entanglement Collapse

(Target Time: ~40 seconds)

> **[Spoken Delivery]**
> *"Thank you, Himmat. Good morning Professors.*

(Point to Figure 5 Plot on Right)
When researchers run projected quantum kernels using default literature settings ($s = 1.00$), the model collapses catastrophically to 55.6% AUC at $N = 10$—barely better than a coin flip! Why?

We identified the exact physical failure mode: the Phase Wrapping Turnaround Bug.
In a 2-layer circuit, rotation angles accumulate across layers ($U(\pi)^2|0\rangle = |0\rangle$). As shown in Figure 5, the Z-axis expectation drops monotonically to -1 at $\theta = \pi/2$, but then turns around and climbs back to $+1$ at $\theta = \pi$, meaning $\langle Z(0)\rangle = +1$ and $\langle Z(\pi)\rangle = +1$.

In clinical terms, this is a disaster: a healthy lung ($\theta \approx 0$) and severe consolidative pneumonia ($\theta \approx \pi$) get mapped to the exact same point in Hilbert space! Compressing the angles ($s \leq 0.25$) confines rotations to the monotonic quadrant $[0, \pi/4]$, which eliminates turnaround and recovers stability to 77.2% AUC."

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slide-11 | Slide 11: Projected Quantum Kernels vs. Fidelity Barren Plateaus

(Target Time: ~40 seconds)

> **[Spoken Delivery]**
> *(Point to Left Box — Fidelity Kernel)*
> *"Next, we address why we used Projected Quantum Kernels instead of standard fidelity kernels.*

A conventional fidelity kernel evaluates global state overlap: $k_{\text{fid}}(\mathbf{x}, \mathbf{x}') = |\langle\phi(\mathbf{x})|\phi(\mathbf{x}')\rangle|^2$. As proved by Huang et al. (Nature Communications 2021), in an n-qubit Hilbert space of dimension 2^n, random states become orthogonal with high probability. The kernel variance concentrates exponentially toward zero: $\text{Var}[k_{\text{fid}}] \sim \mathcal{O}(2^{-n})$. For large n, the Gram matrix collapses into the identity matrix I, destroying margin separability.

(Point to Right Box — Projected Quantum Kernel)
Projected Quantum Kernels bypass this barren plateau by projecting the state back into classical metric space using 1-particle reduced density matrices, measuring 24 local Pauli expectations: $E(\mathbf{x}) \in \mathbb{R}^{24}$.
A classical Gaussian RBF kernel is then evaluated on these expectations. Because the outer function is a standard Gaussian on $\mathbb{R}^{24}$, Mercer's condition is strictly preserved, guaranteeing a positive semi-definite Gram matrix and a convex dual optimization problem."

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slide-12 | Slide 12: Empirical Diagnostics & The Tested ZZ Correlator Hypothesis

(Target Time: ~40 seconds)

> **[Spoken Delivery]**
> *(Point to Figure 4 on Left)*
> *"However, look at our geometric diagnostics in Figure 4. When we measured rank correlation between input Euclidean distance and kernel distance, classical M2 preserved geometry almost perfectly ($r = 0.998$). But under quantum entanglement, the Spearman rank collapsed to $r = 0.412$, scrambling clinical distances! Furthermore, CNOT entangling gates caused single-qubit expectation variance to collapse 34-fold (from 0.290 down to 0.0085).*

(Point to Right Box — ZZ Correlators)
At this stage, we asked a fundamental question:
'Did entanglement fail simply because we measured single qubits individually and threw away two-qubit quantum correlations?'

To test this, we added 8 nearest-neighbor correlators $\langle Z_i Z_{i+1} \rangle$, expanding to 32 total expectations.
The result refuted the hypothesis: test AUC dropped further across all sample sizes (falling by up to -0.034 AUC at $N = 20$). Discarding correlations is not the flaw—multi-qubit entanglement actively corrupts the clinical metric space!

Now, Aryan will present our complete experimental ledger and platform implementation."

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5. Aryan Gupta (2024BCS0020)

Role: 960-Run Ledger, Tuning Budget Accounting, Interactive Streamlit Platform, Bounded Conclusions & Viva Defense
Assigned Slides: 13, 14, 15, 16
Allotted Time: ~2 minutes 05 seconds (~340 words)

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slide-13 | Slide 13: Exact Experimental Accounting: 960 Controlled Runs

(Target Time: ~35 seconds)

> **[Spoken Delivery]**
> *"Thank you, Sawan. Good morning Professors.*

(Point to Left Table — Row Accounting)
To guarantee scientific reproducibility, we executed an exact ledger of 960 controlled runs across 15 seeds:
- 60 runs for Model M1,
- 180 runs for classical Model M2 across feature counts,
- 60 runs for default quantum M3, and
- 660 runs across 11 systematic quantum ablation cells.
Every single run was evaluated on the identical 624-image test set. There was zero cherry-picking and zero missing data.

(Point to Right Box — Budget Asymmetry)
Now, look at the right card: Tuning Budget Asymmetry.
For classical M2, we searched over 36 to 54 hyperparameter configurations per fold. But for quantum M3, we searched over C, γ, AND scale s, evaluating 270 configurations per fold—a $5\times$ larger tuning budget.
Even with this massive search advantage skewed in quantum's favor, quantum failed to outperform the classical control."

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slide-14 | Slide 14: Empirical Benchmark Ledger & Bonferroni Significance

(Target Time: ~35 seconds)

> **[Spoken Delivery]**
> *(Point to Table — Benchmark Results)*
> *"Here is the definitive empirical ledger across 15 seeds ($df = 14$):*
> *At $N = 10$, classical M2 reaches 0.8957 AUC, while tuned quantum M3 reaches 0.7717 AUC—a significant deficit of -0.1241 ($p = 2.4 \times 10^{-4}$).*
> *At $N = 20$, the deficit is -0.0846 ($p = 3.8 \times 10^{-4}$).*
> *At $N = 50$, the deficit is -0.0296 ($p = 7.9 \times 10^{-4}$).*
> *At $N = 100$, the deficit is -0.0288 ($p = 1.5 \times 10^{-4}$).*

Every single paired difference survives the Bonferroni threshold ($\alpha = 0.00417$). The hypothesis of quantum advantage on scarce radiographs is conclusively rejected."

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slide-15 | Slide 15: Streamlit Research Prototype & Disclaimer

(Target Time: ~30 seconds)

> **[Spoken Delivery]**
> *(Point to Prototype Architecture on Left)*
> *"To make our benchmark interactive, we engineered a production platform in Streamlit. Instead of running expensive quantum simulations during inference, we pre-compiled a lightweight 28.6 MB internal database (`prototype_db.npz`). This enables instant switching across all 624 patients, seeds, and circuits in under 5 milliseconds.*

It is deployed live on Oracle Cloud under Caddy with verified Let's Encrypt SSL at `qkernel-lite.duckdns.org`, passing all 32 unit tests in 28 seconds.

(Point to Disclaimer on Right)
Crucially, we enforce a mandatory clinical disclaimer: this is strictly an academic benchmark and not a certified medical device."

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slide-16 | Slide 16: Bounded Conclusions & Viva Defense Integrity

(Target Time: ~25 seconds)

> **[Spoken Delivery]**
> *(Point to Permissible Claims on Left)*
> *"In conclusion, we stand firmly behind four bounded claims:*
> *1. Pretrained transfer learning accounts for 100% of diagnostic capability.*
> *2. Projected quantum kernels remain statistically inferior to matched classical RBF-SVM across all sample sizes.*
> *3. The only quantum circuit matching classical performance is a 1-layer product state, which is classically simulable.*
> *4. Multi-qubit entanglement actively degrades diagnostic metric geometry.*

(Point to Bottom Card)
We do not claim quantum kernels can never work on images—we report what we measured with complete statistical rigor. In engineering science, a reproducible null result is vastly superior to an unreproducible claim of advantage.

Thank you, Professors. We now welcome your questions."

(All five presenters stand together, ready for viva questions).